Water Cycle and Hydrology

The water cycle describes how water moves among oceans, atmosphere, land, rivers, groundwater, snow, and ice.

Key facts:

Evaporation changes liquid water into water vapor, mainly from oceans, lakes, rivers, and wet soils.

Transpiration is water vapor released by plants through stomata; together with evaporation it is called evapotranspiration.

Condensation forms cloud droplets when moist air cools to its dew point.

Precipitation includes rain, snow, sleet, and hail that fall from clouds to Earth’s surface.

Infiltration moves water from the surface into soil, while percolation moves it deeper toward groundwater.

Runoff flows over land into streams and rivers when rainfall exceeds infiltration or soil storage.

Groundwater is stored in aquifers and can discharge naturally into springs, rivers, lakes, and oceans.

Answerable questions:

Q: What drives the water cycle? A: Solar energy and gravity drive evaporation, atmospheric movement, precipitation, runoff, and groundwater flow.

Q: Why is infiltration important? A: It recharges soil moisture and groundwater and reduces the amount of immediate surface runoff.